Edward Silva

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Education

Colorado School of Mines May 2026

BS Electrical Engineering, Minor in Computer Science, Golden, CO, GPA: 3.44/4.00

Honors: Dean's List (Fall 2022, Fall 2024, Spring 2026) | Honor Roll (Fall 2023, Spring 2024, Fall 2025)

Scholarships: Provost Scholarship | 2023–2024 Computing-Mines Affiliates Partnership Program (C-MAPP) Scholar (Frontier Technology Inc.) | 2024–2025 C-MAPP Scholar (Parsons Corporation) | 2024–2025 American Bureau of Shipping Scholar

Graduate: Estimation Theory & Kalman Filtering (A)

Relevant: Digital Signal Processing, Modern Control Design, Feedback Control Systems, Embedded Systems, Engineering Electromagnetics, Elements of Computing Systems, Printed Circuit Board Design

Skills

Languages: C, C++, Python, MATLAB
Tools: Simulink, Git, Linux, Intel VTune, Jupyter, OpenCV, Arduino, Autodesk Revit, AutoCAD MEP
Methods: Kalman filtering (EKF, UKF), LQR and observer design, SIMD (AVX-512), FIR filters, I2C

Experience

Embedded Software Engineer, SpaceX Jul 2026 – Present
C, C++, Linux, Git Bastrop, TX

- Write embedded C and C++ software and Linux kernel drivers for Starlink phased array systems, interfacing with the RF, silicon, and network software teams
- Cut the run time of a calculation in the Starlink phased array software by 11%

Software Engineering Intern, Kratos Defense Jun – Aug 2025
C++, Python, SIMD, Jupyter, Git Colorado Springs, CO

- Sped up legacy real-time C++ DSP code by 1.6x with refactoring and AVX-512 vectorization, guided by VTune
- Analyzed code timings in Python and created Jupyter plots of the results
- Presented a faster FIR filter implementation and its performance trade-offs to the team for use on later projects
- Built a configurable logging framework with severity levels and error tracking for the real-time DSP software

Co-op Intern, Electrical Design, Jordan and Skala Engineers Jan – Jun 2025
Autodesk Revit, AutoCAD MEP, NEC Compliance Denver, CO

- Drafted layouts, panel schedules, and riser diagrams for 20+ apartment and early data center projects, with NEC load and voltage-drop calculations

Teaching

Calculus II and Differential Equations Grader, Colorado School of Mines, Remote Jan – Aug 2024

- Graded Calculus II and Differential Equations for 30–50 students a semester across 1–2 classes

Peer Mentor, Colorado School of Mines, Golden, CO Aug – Dec 2023

- Co-taught a 30-student section of the one-credit First Year Success Seminar and led an orientation group

Projects

Electromagnetic Bicycle Damper (EMBR), MATLAB, Python — Fall 2025 – Spring 2026

- Owned the electromagnet design and calculation verification within an 8-person interdisciplinary team, sponsored by ARUS MR Tech, that retrofitted a production front suspension fork with a semi-active MR damper
- Modeled a ∼250-turn, 2 A coil in Python by summing Biot-Savart loop fields with complete elliptic integrals, estimating a 0.15 T peak across the 2.5 mm radial gap
- Designed the flux path through the steel parts so the field crosses the annular gap perpendicular to the fluid flow

Nonlinear State Estimation for the EMBR Damper (graduate course, solo), MATLAB, Simulink — Spring 2026

- Modeled the MR damper, MOSFET-driven coil, and sensor array as a 4-state nonlinear system in MATLAB and Simulink, adding terrain force to the state as a random walk
- Implemented and compared extended (EKF) and unscented (UKF) Kalman filters to estimate unmeasured states
- Showed the UKF cut RMS force error from 624 N to 401 N at a larger yield constant, keeping 100% of samples inside 3σ against 58% for the EKF

Pitch-Tracking Systems in Auto-Tune, MATLAB — Spring 2026

- Pair-programmed real-time FFT and autocorrelation (ACF) pitch detectors in MATLAB with a teammate, and found in a 50 Hz to 4 kHz chirp sweep that the ACF failed near 1600 Hz and the FFT near 1800 Hz
- Measured mean per-frame latency of 0.22 ms for the FFT against 0.59 ms for the ACF (2.68x), with 0.26 ms against 0.74 ms standard deviation

Gantry Crane Control System, MATLAB, Simulink — Fall 2025

- Designed a 50 Hz discrete-time gantry-crane controller in the z-domain in a 3-person team, with 0% overshoot, 112.6 s settling against a 120 s spec, and cart velocity ≤ 1 m/s
- Found a 48° phase margin and 0.52 s delay tolerance, confirmed by injecting a 25-sample delay
- Pair-programmed an LQR controller with a full-order observer, comparing reference scaling (3.95 s, 9.02% overshoot) with integral augmentation plus anti-windup (3.45 s, negligible overshoot)

Autonomous Maze-Navigating Robot, Python — Fall 2025

- Achieved the second-fastest course completion time in the cohort with an autonomous maze-navigating robot built by a 4-person team, integrating computer vision, embedded control, and I2C communication
- Built the Python and OpenCV ArUco vision subsystem with a teammate, splitting its 640x480 pipeline into capture, processing, display, and LCD threads so the Arduino stopped waiting on the vision code
- Fixed an I2C packet mismatch by repacking 8-byte Python doubles as 4-byte little-endian floats for the Arduino

eeprep (eeprep.com) — Ongoing

- Wrote a free CC BY 4.0 job guide for EECS students and new graduates, with 30 pages and 350+ linked sources

Certifications

MATLAB Machine Learning Techniques, MathWorks

Organizations

IEEE-HKN, Nu Pi Chapter, Colorado School of Mines, Member — 2026 – Present
IEEE, Member — 2025 – Present
IEEE Young Professionals, Member — 2025 – Present